Asteroseismology and the Radial Adiabatic Pulsation Equations

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Objectives

- Motivate the derivation of radial adiabatic pulsation
- Set up the equations from fluid mechanics applied to stellar p, ρ, and Φ gradients
- Perturb these equations to the first order
- Explain the radial adiabatic solution process
- Solve for a toy sun with uniform density

What is Asteroseismology?

Asteroseismology is the **study of oscillations in stars**. Its model of these oscillations helps us **make predictions about internal stellar structure from observed oscillations**, and vice versa.

The radial adiabatic model is a useful, simple model of stellar oscillations. It asks us to consider: **what happens if we radially perturb a star in equilibrium while conserving entropy?**

We use more complex stellar models that allow for non-radial, non-adiabatic pulsation to model variable stars like Cepheids and δ Scutis. These are solved using numerical methods for the equations of motion, and provide **independent constraints on stellar properties**.

Stellar Equations at Equilibrium

The adiabatic radial model comes from the **mass**, **momentum**, and **energy conservation** equations, subject to **gravity** and **adiabaticity**.

$$\text{Mass: } \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \tag{1}$$

$$\text{Momentum: } \rho_0 \frac{\partial v}{\partial t} = -\frac{\partial p_0}{\partial r} - \rho_0 \frac{\partial \Phi}{\partial r} = \underset{\text{eq}}{0} \tag{2}$$

$$\text{Energy: } \frac{\mathrm{d}p}{\mathrm{d}t} = \frac{\Gamma_1 p}{\rho}\frac{\mathrm{d}\rho}{\mathrm{d}t}, \quad \Gamma_1 = \left(\frac{\partial \ln p}{\partial \ln \rho}\right)_{\text{s}} \tag{3}$$

$$\text{Gravity: } f_{\text{g}} = -\rho_0 \frac{\partial \Phi}{\partial r}, \quad \nabla^2 \Phi = 4\pi G \rho_0 \tag{4}$$

Perturbation Analysis

We can derive solutions for the radial adiabatic pulsation by starting with a **radial Lagrangian perturbation** of δr. Moving a parcel of gas from r_0 to $r_0 + \delta r$, the Lagrangian δp of the parcel is:

$$\underbrace{\delta p(r)}_{\text{Lagrangian}} = p(r_0 + \delta \mathbf{r}) - p_0(\mathbf{r_0}) =$$
$$p(\mathbf{r_0}) + \delta \mathbf{r} \cdot \nabla p_0 - p_0(\mathbf{r_0}) = \underbrace{p'(\mathbf{r_0})}_{\text{Eulerian}} + \delta \mathbf{r} \frac{\mathrm{d}p_0}{\mathrm{d}r} \tag{5}$$

This is the **Eulerian** perturbation p' plus the radial perturbation of the pressure gradient $\frac{\mathrm{d}p_0}{\mathrm{d}r}$.

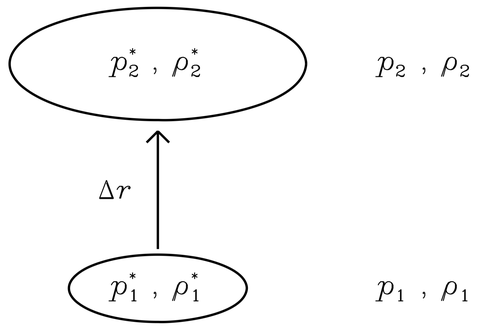


Figure 1: Consider a parcel going from $\mathbf{r_1}$ to $\mathbf{r_2}$, where $*$ denotes the new state. $p_2^* - p_1$ is δp; $p_2^* - p_2$ is p'. Figure from [1].

Boundary Conditions

Eq. 8 relates eigenfrequency ω^2 and eigenfunction $\zeta(r,t)$ to the equilibrium profiles r_0, p_0, ρ_0, and Γ_0. For the adiabatic radial case, **this is tractable** with boundary conditions: a perturbation at $r = 0$ is linear with radius, and $r = R$ is a free surface.

$$\lim_{r \to 0} \zeta(r) = \text{const} \tag{6}$$

$$\delta p(R) = p'(R) + \xi_r(R) \left.\frac{\mathrm{d}p_0}{\mathrm{d}r}\right|_R = 0. \tag{7}$$

Solving for the Sun

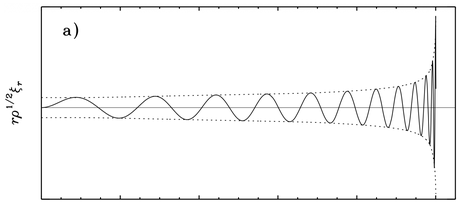


Figure 2: A high-order ($n = 23$) radial Solar eigenfunction from $r = [0, R_\odot]$ at $\nu = 3310$ μHz (5 mins). Figure from [1].

Using equilibrium profiles and Sturm-Liouville theory, we can solve for the eigenfrequencies in Eq. 8. The wave above is a high-frequency mode; the y axis $y = r\rho^{1/2}\xi_r$ is the square root of the integrand of the mode inertia ε, where average energy $\bar{E} = 1/4\omega^2\varepsilon$.

Adiabatic Radial Pulsation Equation

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r_0^4 \Gamma_{1,0} p_0 \frac{\mathrm{d}\zeta}{\mathrm{d}r}\right) + r_0^3 \zeta \frac{\mathrm{d}}{\mathrm{d}r}\left[(3\Gamma_{1,0} - 4)p_0\right] + \rho_0 \omega^2 r_0^4 \zeta = 0, \quad \zeta(r) = \frac{\xi_r(r)}{r}. \tag{8}$$

Perturbed Stellar Equations

$$\text{Mass: } \frac{\partial \rho'}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \rho_0 v\right) = 0. \tag{9}$$

$$\text{Momentum: } \rho_0 \frac{\partial^2 \xi_r}{\partial t^2} = -\frac{\partial p'}{\partial r} - \rho' g_0 - \rho_0 \frac{\partial \Phi'}{\partial r} \tag{10}$$

$$\text{Energy: } p' + \xi \frac{\mathrm{d}p}{\mathrm{d}r} = \frac{\Gamma_{1,0} p_0}{\rho_0}\left(\rho' + \xi \frac{\mathrm{d}\rho}{\mathrm{d}r}\right) \tag{11}$$

$$\text{Gravity: } \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \Phi'}{\partial r}\right) = 4\pi G \rho' \tag{12}$$

Simplified Uniform ρ Model

To solve for ω^2 without solving eigenfunctions, we can assume $\frac{\mathrm{d}\zeta}{\mathrm{d}r} = 0$. Further assuming uniform ρ, this is a star in **simple harmonic motion**:

$$\frac{\mathrm{d}}{\mathrm{d}r}[(3\Gamma_1 - 4)p] = -\rho r \omega^2 \tag{13}$$

Using $\frac{\mathrm{d}p}{\mathrm{d}r} = -\rho g(r)$ and $g(r) = Gm(r)/r^2$:

$$\omega^2 = -\frac{\mathrm{d}}{\mathrm{d}r}(3\Gamma_1 - 4)\frac{p}{\rho r} = (3\Gamma_1 - 4)\frac{Gm(r)}{r^3} \tag{14}$$

For $\Gamma_1 = 5/3$ (ideal H gas), $\omega^2 = Gm(r)/r^3$, giving stable oscillation with $\boxed{\text{P} = 2\pi\sqrt{r^3/Gm(r)}}$.

Checking with a Toy Sun ⊙

With $M = M_\odot$, $R = R_\odot$, we get $P = 10\,010\,\text{s}$. This is exactly equivalent to P_{orb} at the star's surface, given by the equation $P_{\text{orb}} = 2\pi\sqrt{R^3/GM_\odot}$. The **dynamical timescales of oscillation and surface orbit are equal** for this model.

Assumptions and Simplifications

- No rotation, no coulomb forces or viscosity
- Spherical symmetry of geometry and $\rho, p,$ and Φ distributions
- Non-adiabatic (heat transfer) effects can be ignored due to long timescales if ρ isn't tiny
- $\tau_\epsilon \gg P_{\text{puls}} \implies$ nuclear reactions can be ignored.
- Magnetic fields can be ignored; no MHD effects
- Pulsation heat transfer per cycle is negligible
- The exterior of the star at $r = R_\star$ is a free surface
- The star is a continuum where each $f(r)$ is defined for every $r \in [0, R_\star]$

References

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Acknowledgements

I'm very grateful for Aerts, Kurtz, Kawaler, Christensen-Dalsgaard, Knapp, and so many others who have both pioneered the techniques of asteroseismology and endeavored to share them in an accessible manner. I'm also quite thankful for the guidance of Dr. José Zorrilla throughout.

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